

Product, Quotient, and Trigonometric Derivatives



The product rule

- 1 Differentiate $f(x) = (3x^2 - 1)(x^3 + 2x)$ using the product rule.
 - 2 Differentiate $f(x) = x^2 \sin x$.
 - 3 Differentiate $f(x) = e^x \cos x$.
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The quotient rule

- 4 Differentiate $f(x) = \frac{3x-1}{x+4}$.
 - 5 Differentiate $f(x) = \frac{x^2}{x^2+1}$.
 - 6 Differentiate $f(x) = \frac{\sin x}{x}$.
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Derivatives of the six trig functions

- 7 State the derivative of each trig function:

a $\frac{d}{dx} \sin x$	b $\frac{d}{dx} \cos x$	c $\frac{d}{dx} \tan x$
d $\frac{d}{dx} \cot x$	e $\frac{d}{dx} \sec x$	f $\frac{d}{dx} \csc x$
 - 8 Differentiate $f(x) = x \tan x$.
 - 9 Differentiate $f(x) = \sec x \cdot \csc x$.
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Combining product, quotient, and trig rules

- 10 Differentiate $f(x) = \frac{e^x}{\sin x}$.
 - 11 Find the equation of the tangent line to $f(x) = x \cos x$ at $x = 0$.
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Derivatives using a table of values

- 12 Given $f(3) = 2$, $f'(3) = -1$, $g(3) = 4$, $g'(3) = 5$, find $h'(3)$ if $h(x) = f(x)g(x)$.

- 13 Using the same table, find $k'(3)$ if $k(x) = \frac{f(x)}{g(x)}$.
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Second derivatives with product and quotient rules

- 14 Find $f''(x)$ for $f(x) = x \sin x$.
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More product and quotient rule practice

- 15 Differentiate $f(x) = (2x + 1)^2(x - 3)$, treating $(2x + 1)^2$ as one factor.
- 16 Differentiate $f(x) = \frac{x^2 + 1}{x - 1}$.
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An applied product-rule problem

- 17 A company's revenue is $R(t) = p(t) \cdot q(t)$, where $p(t)$ is price and $q(t)$ is quantity sold, both functions of time t (months). At $t = 6$, $p(6) = 20$, $p'(6) = 0.5$, $q(6) = 100$, $q'(6) = -2$. Find $R'(6)$ and interpret its sign.
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Higher-order trigonometric derivatives

- 18 Find $f''(x)$ for $f(x) = \sin x$, and describe the repeating pattern of derivatives f, f', f'', f''', f'''' .
- 19 Using the repeating pattern above, find $f^{(50)}(x)$ for $f(x) = \sin x$ (the 50th derivative).
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A capstone combined-rule problem

- 20 Differentiate $f(x) = \frac{x^2 \sin x}{x + 1}$, applying the quotient rule with the product rule for the numerator's derivative.
- 21 An object's position is $s(t) = t^2 \cos t$ (meters, seconds). Find the object's velocity function $s'(t)$ using the product rule, then find its velocity at $t = 0$.
- 22 Differentiate $f(x) = \frac{\cos x}{1 + \sin x}$, and simplify your answer using the Pythagorean identity $\sin^2 x + \cos^2 x = 1$.

- 23 Given $f(2) = 5$, $f'(2) = 3$, $g(2) = -1$, $g'(2) = 4$, find each of the following at $x = 2$:
- a $\frac{d}{dx}[f(x)g(x)]$
 - b $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right]$
 - c $\frac{d}{dx}[3f(x) - 2g(x)]$
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Multiple choice — easy 15

- 24 Given $f(2) = 3$, $f'(2) = 5$, $g(2) = 4$, $g'(2) = -1$, find $h'(2)$ if $h(x) = f(x)g(x)$.
- a. -5
 - b. 23
 - c. 17
 - d. 12
- 25 Given $f(7) = 4$, $f'(7) = -4$, $g(7) = 6$, $g'(7) = -2$, find $h'(7)$ if $h(x) = f(x)g(x)$.
- a. -16
 - b. 24
 - c. 8
 - d. -32
- 26 Given $f(6) = -5$, $f'(6) = 1$, $g(6) = 1$, $g'(6) = -1$, find $h'(6)$ if $h(x) = f(x)g(x)$.
- a. -4
 - b. 6
 - c. -1
 - d. -5

- 27 Given $f(4) = 5$, $f'(4) = -4$, $g(4) = 1$, $g'(4) = 5$, find $h'(4)$ if $h(x) = f(x)g(x)$.
- a. -20
 - b. 5
 - c. -29
 - d. 21
- 28 Given $f(4) = -3$, $f'(4) = 3$, $g(4) = -5$, $g'(4) = 1$, find $h'(4)$ if $h(x) = f(x)g(x)$.
- a. -12
 - b. 15
 - c. 3
 - d. -18
- 29 Given $f(5) = 3$, $f'(5) = 4$, $g(5) = -6$, $g'(5) = -1$, find $h'(5)$ if $h(x) = f(x)g(x)$.
- a. -4
 - b. -21
 - c. -18
 - d. -27
- 30 Given $f(3) = -4$, $f'(3) = 6$, $g(3) = 3$, $g'(3) = -4$, find $h'(3)$ if $h(x) = f(x)g(x)$.
- a. -24
 - b. -12
 - c. 2
 - d. 34

- 31 Given $f(7) = 4$, $f'(7) = -2$, $g(7) = -5$, $g'(7) = 1$, find $h'(7)$ if $h(x) = f(x)g(x)$.
- a. 14
 - b. -2
 - c. -20
 - d. 6
- 32 Given $f(3) = 2$, $f'(3) = 4$, $g(3) = -5$, $g'(3) = 3$, find $h'(3)$ if $h(x) = f(x)g(x)$.
- a. -10
 - b. -14
 - c. -26
 - d. 12
- 33 Given $f(5) = 2$, $f'(5) = -2$, $g(5) = 5$, $g'(5) = -2$, find $h'(5)$ if $h(x) = f(x)g(x)$.
- a. -14
 - b. 4
 - c. 10
 - d. -6
- 34 Given $f(6) = 3$, $f'(6) = -1$, $g(6) = -6$, $g'(6) = 1$, find $h'(6)$ if $h(x) = f(x)g(x)$.
- a. -1
 - b. 9
 - c. 3
 - d. -18

- 35 Given $f(4) = 6$, $f'(4) = -4$, $g(4) = 6$, $g'(4) = 5$, find $h'(4)$ if $h(x) = f(x)g(x)$.
- a. -54
 - b. -20
 - c. 36
 - d. 6
- 36 Given $f(8) = 1$, $f'(8) = -3$, $g(8) = -3$, $g'(8) = -5$, find $h'(8)$ if $h(x) = f(x)g(x)$.
- a. 14
 - b. 4
 - c. 15
 - d. -3
- 37 Given $f(3) = -2$, $f'(3) = -3$, $g(3) = -4$, $g'(3) = -1$, find $h'(3)$ if $h(x) = f(x)g(x)$.
- a. 3
 - b. 10
 - c. 14
 - d. 8
- 38 Given $f(4) = -5$, $f'(4) = 1$, $g(4) = 4$, $g'(4) = 6$, find $h'(4)$ if $h(x) = f(x)g(x)$.
- a. -26
 - b. 34
 - c. -20
 - d. 6

Multiple choice — medium 15

39 Given $f(2) = -5$, $f'(2) = -4$, $g(2) = 5$, $g'(2) = -4$, find $k'(2)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. $\frac{8}{5}$

b. -8

c. 0

d. $-\frac{8}{5}$

40 Given $f(4) = -4$, $f'(4) = -1$, $g(4) = 6$, $g'(4) = -2$, find $k'(4)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. $\frac{1}{18}$

b. $\frac{7}{18}$

c. $-\frac{7}{18}$

d. $-\frac{7}{3}$

41 Given $f(2) = -1$, $f'(2) = 6$, $g(2) = -3$, $g'(2) = 4$, find $k'(2)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. $\frac{14}{3}$

b. $-\frac{22}{9}$

c. $-\frac{14}{9}$

d. $\frac{14}{9}$

42 Given $f(2) = 6$, $f'(2) = -1$, $g(2) = 5$, $g'(2) = -6$, find $k'(2)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. $\frac{31}{25}$

b. $-\frac{41}{25}$

c. $-\frac{31}{25}$

d. $\frac{31}{5}$

43 Given $f(4) = -2$, $f'(4) = 2$, $g(4) = -3$, $g'(4) = -1$, find $k'(4)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. $-\frac{4}{9}$

b. $\frac{8}{9}$

c. $\frac{8}{3}$

d. $-\frac{8}{9}$

44 Given $f(2) = -1$, $f'(2) = -1$, $g(2) = -2$, $g'(2) = 2$, find $k'(2)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. -2

b. -1

c. 0

d. 1

45 Given $f(5) = 2$, $f'(5) = 2$, $g(5) = 2$, $g'(5) = -3$, find $k'(5)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. $-\frac{5}{2}$

b. $\frac{5}{2}$

c. $-\frac{1}{2}$

d. 5

46 Given $f(4) = -6$, $f'(4) = 4$, $g(4) = -4$, $g'(4) = 2$, find $k'(4)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. $\frac{1}{4}$

b. $-\frac{7}{4}$

c. $-\frac{1}{4}$

d. 1

47 Given $f(2) = -3$, $f'(2) = 4$, $g(2) = 2$, $g'(2) = 4$, find $k'(2)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. -1

b. -5

c. 5

d. 10

48 Given $f(3) = 4$, $f'(3) = 5$, $g(3) = 3$, $g'(3) = -1$, find $k'(3)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. $-\frac{19}{9}$

b. $\frac{19}{3}$

c. $\frac{19}{9}$

d. $\frac{11}{9}$

49 Given $f(8) = -2$, $f'(8) = -5$, $g(8) = 4$, $g'(8) = -5$, find $k'(8)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. $-\frac{15}{2}$

b. $-\frac{15}{8}$

c. $-\frac{5}{8}$

d. $\frac{15}{8}$

50 Given $f(3) = -6$, $f'(3) = 2$, $g(3) = -4$, $g'(3) = 1$, find $k'(3)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. $-\frac{7}{8}$

b. $\frac{1}{8}$

c. $-\frac{1}{8}$

d. $\frac{1}{2}$

51 Given $f(6) = 3$, $f'(6) = -2$, $g(6) = 4$, $g'(6) = 6$, find $k'(6)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. $\frac{5}{8}$

b. $-\frac{13}{2}$

c. $\frac{13}{8}$

d. $-\frac{13}{8}$

52 Given $f(3) = -6$, $f'(3) = -6$, $g(3) = 6$, $g'(3) = -3$, find $k'(3)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. -9

b. $\frac{3}{2}$

c. $-\frac{3}{2}$

d. $-\frac{1}{2}$

53 Given $f(8) = -5$, $f'(8) = 5$, $g(8) = 4$, $g'(8) = 6$, find $k'(8)$ if $k(x) = \frac{f(x)}{g(x)}$.

a. $\frac{25}{2}$

b. $\frac{25}{8}$

c. $-\frac{5}{8}$

d. $-\frac{25}{8}$

Multiple choice — hard 15

54 Use the product rule to differentiate $f(x) = (2x - 5)(4x - 4)$, then find $f'(1)$.

a. -12

b. 0

c. 8

d. -10

- 55 Use the product rule to differentiate $f(x) = (4x - 2)(5x - 4)$, then find $f'(1)$.
- a. 2
 - b. 16
 - c. 14
 - d. 20
- 56 Use the product rule to differentiate $f(x) = (5x - 3)(4x + 2)$, then find $f'(-1)$.
- a. -40
 - b. -42
 - c. 20
 - d. 16
- 57 Use the product rule to differentiate $f(x) = (x - 2)(2x - 3)$, then find $f'(3)$.
- a. 3
 - b. 5
 - c. 7
 - d. 2
- 58 Use the product rule to differentiate $f(x) = (2x + 1)(2x + 6)$, then find $f'(-3)$.
- a. -8
 - b. -10
 - c. 0
 - d. 4

59 Use the quotient rule to differentiate $f(x) = \frac{3x+3}{3x+6}$, then find $f'(0)$.

a. $\frac{1}{4}$

b. $-\frac{1}{4}$

c. $\frac{5}{4}$

d. $\frac{3}{2}$

60 Use the quotient rule to differentiate $f(x) = \frac{2x-6}{2x-3}$, then find $f'(0)$.

a. -2

b. $\frac{1}{6}$

c. $-\frac{2}{3}$

d. $\frac{2}{3}$

61 Use the quotient rule to differentiate $f(x) = \frac{2x-6}{3x+6}$, then find $f'(1)$.

a. $\frac{10}{27}$

b. $-\frac{7}{54}$

c. $\frac{10}{3}$

d. $-\frac{10}{27}$

62 Use the quotient rule to differentiate $f(x) = \frac{3x-1}{2x+1}$, then find $f'(2)$.

a. $\frac{1}{5}$

b. 1

c. $-\frac{1}{5}$

d. $\frac{6}{5}$

- 63 Use the quotient rule to differentiate $f(x) = \frac{3x-5}{3x-1}$, then find $f'(2)$.
- a. $\frac{37}{25}$
 - b. $\frac{12}{5}$
 - c. $-\frac{12}{25}$
 - d. $\frac{12}{25}$
- 64 Use the product rule to differentiate $f(x) = -7x\cos x$, then find $f'(0)$.
- a. -8
 - b. 7
 - c. -6
 - d. -7
- 65 Use the product rule to differentiate $f(x) = -4x\cos x$, then find $f'(0)$.
- a. 4
 - b. -5
 - c. -4
 - d. -3
- 66 Use the quotient rule to differentiate $f(x) = \frac{\sin x}{x+2}$, then find $f'(0)$.
- a. $-\frac{1}{2}$
 - b. $\frac{1}{3}$
 - c. 2
 - d. $\frac{1}{2}$

67 Use the quotient rule to differentiate $f(x) = \frac{\sin x}{x-2}$, then find $f'(0)$.

a. -1

b. $\frac{1}{2}$

c. $-\frac{1}{2}$

d. -2

68 Use the quotient rule to differentiate $f(x) = \frac{\sin x}{x-1}$, then find $f'(0)$.

a. $\frac{1}{2}$

b. 0

c. 1

d. -1